

Mingxuan Jin

PhD Researcher | Software Development and Interactive Systems

Cardiff, UK | mingxuanj269@gmail.com | +44 7784 600992

EDUCATION

Cardiff University - PhD, Computer Science & Informatics School of Computational and Mathematical Sciences; expected completion Apr 2028	Apr 2024 - present
University of Nottingham Master's, Human-Computer Interaction (HCI) (Distinction)	Sep 2021 - Sep 2022
Beijing University of Technology & University College Dublin Bachelor's, Software Engineering	Sep 2016 - Jul 2020

INDUSTRY EXPERIENCE

Matrixtime Robotics Co., Ltd. <i>Deep Learning Technology Engineer Intern / Shanghai, China</i> Developed Python CNN-based vision algorithms for robotic defect inspection of automotive components; preprocessed images and analysed inspection data.	Mar 2023 - Jul 2023
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SELECTED RESEARCH AND ENGINEERING

Cardiff University, PhD Researcher (Apr 2024 - present)

AR Walking System with Sensor and API Integration <i>Developer / Magic Leap 2, Python; ACM Conference on Human Factors in Computing Systems 2027 under review</i> M. Zhou, Mingxuan Jin, K. Stawarz, X. Peng, A. Denisova, H. Egami, T. Wen, Y. Gu, D. J. Finnegan, S. C. Barathi. - Built the sensor-data pipeline and movement classifier, integrating IMU cues with eight human-confirmed route levels and distance-based AR guide behaviour. - Implemented structured API inputs, action-consistency validation and deterministic fallback so generated wording communicated the fixed progression decision.	Mar 2026 - Sep 2026
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Children's VR Cycling Interaction and Logging System <i>Lead developer and evaluator / Unity/C#; ACM Conference on Human Factors in Computing Systems 2027 under review</i> Mingxuan Jin, M. Zhou, X. Peng, D. J. Finnegan, L. D. Turner, S. C. Barathi. Connected cycling input, automatic movement and gesture dialogue; implemented four controlled event sequences and synchronised power/event logging, evaluated in studies with 16 and 68 children.	Mar 2026 - Sep 2026
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Configurable VR Cycling Platform for Older Adults <i>Lead developer and evaluator / International Journal of Human-Computer Studies under review after minor revisions</i> M. Zhou, Mingxuan Jin (co-first author), X. Peng, K. Stawarz, A. Denisova, J. Huang, S. C. Barathi. Built bike-controlled navigation, branching dialogue and six experimental configurations; led deployment and evaluation in two studies ($n = 15$ and $n = 36$).	May 2025 - Sep 2025
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SELECTED SOFTWARE PROJECTS

Bot Detection Web App with Model Explanations <i>Python, Django, JavaScript</i> Built a Django application with a Random Forest classifier using 12 selected features, Bayesian hyperparameter optimisation, and visual/natural-language explanations from decision trees.	Feb 2020 - Jun 2020
NLP Review Classification and Editing Web App <i>Python, JavaScript, HTML/CSS</i> Combined Naive Bayes-derived aspect features with an SVM rating classifier; built rule-based editing suggestions and a web interface, then iterated through usability testing.	May 2022 - Sep 2022

CORE SKILLS

Languages: Python, C#, Java, JavaScript, SQL **Tools:** Unity, Django, MySQL, Git/GitHub

Systems: Bluetooth/IMU integration, external APIs, event logging **ML:** Random Forest, CNN-based vision, SVM